

BEAMS WITHOUT LATERAL SUPPORT

The allowable compressive stress due to bending about the x-x axis for universal beams and columns, joists and channels is the lesser of the values of p_{bc} given in Tables 2 and 3 a, b or c (depending on the grade of steel) in BS 449. The values in Table 2 apply when the compression flange is so supported that lateral instability is obviated. When the compression flange is unsupported or insufficiently supported laterally, the appropriate Table 3 applies. The maximum effective length of compression flange which may be used without reduction of the allowable bending stresses in Table 2 is given as L_c with the safe load tables.

Example: Find the allowable stress, p_{bc} and the safe uniformly distributed load, W , for a 533 = 210 UB 82 in grade 43 steel spanning 7m without lateral restraint and having the ends of the compression flange partially restrained.

Effective length = $0.85 \times 700 = 595$ cm (see BS 449, Clause 26)

$l/r_y 595/4.38 = 136$; $D/T = 40$.

From BS 449, Table 3a, $p_{bc} = 119 \text{ N/mm}^2$

The safe uniformly distributed load, W

$$\begin{aligned}
 &= 8fZ/L \\
 &= (8 \times 119 \times 1798 \times 10^3 / 7000) \text{ N} \\
 &= 245 \text{ kN}
 \end{aligned}$$

Lateral instability is not a criterion for these sections when bent about the y-y axis and the allowable stresses in Table 2 apply.

